

Teacher Answer Key:

Mechanical Oscillations – Ex 7: Oscillator Between Two Springs

1. Equilibrium Elongations ($\Delta l_0, \Delta l'_0$)

Conditions:

1. Geometric Condition: The total length $AB = 50$ cm. The sum of free lengths is $2l_0 = 40$ cm.

$$\Delta l_{total} = AB - 2l_0 = 50 - 40 = 10 \text{ cm.}$$

$$\Delta l_0 + \Delta l'_0 = 0.1 \text{ m} \quad \text{--- (1)}$$

2. Equilibrium Condition: The tensions balance each other ($T = T'$).

$$k\Delta l_0 = k'\Delta l'_0$$

$$8\Delta l_0 = 12\Delta l'_0 \implies \Delta l_0 = 1.5\Delta l'_0 \quad \text{--- (2)}$$

Substituting (2) into (1):

$$1.5\Delta l'_0 + \Delta l'_0 = 0.1 \implies 2.5\Delta l'_0 = 0.1$$

$$\Delta l'_0 = \frac{0.1}{2.5} = 0.04 \text{ m.}$$

$$\Delta l_0 = 0.1 - 0.04 = 0.06 \text{ m.}$$

Results: $\Delta l_0 = 6 \text{ cm}, \quad \Delta l'_0 = 4 \text{ cm}$

2. Potential Energy at Equilibrium

The gravitational potential energy is zero. The system only has Elastic Potential Energy (EPE).

$$PE_{eq} = EPE_{(r)} + EPE_{(r')}$$

$$PE_{eq} = \frac{1}{2}k\Delta l_0^2 + \frac{1}{2}k'\Delta l_0'^2$$

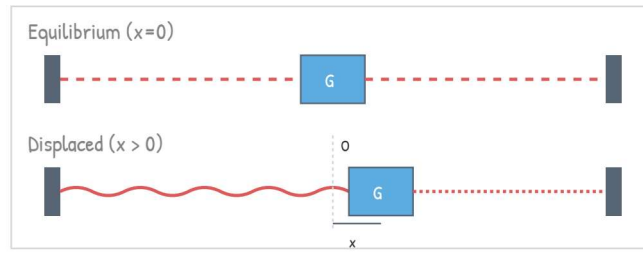
$$PE_{eq} = \frac{1}{2}(8)(0.06)^2 + \frac{1}{2}(12)(0.04)^2$$

$$PE_{eq} = 4(0.0036) + 6(0.0016) = 0.0144 + 0.0096$$

Result: $PE_{eq} = 0.024 \text{ J}$

3. Dynamics of the Motion

a) Construction of Figures:



b) Mechanical Energy Expression:

When displaced by x to the right:

- Elongation of (r) becomes: $\Delta l = \Delta l_0 + x$.
- Elongation of (r') becomes: $\Delta l' = \Delta l'_0 - x$.

$$ME = KE + EPE_{tot} = \frac{1}{2}Mv^2 + \frac{1}{2}k(\Delta l_0 + x)^2 + \frac{1}{2}k'(\Delta l'_0 - x)^2$$

Expanding the squares:

$$\frac{1}{2}k(\Delta l_0^2 + 2x\Delta l_0 + x^2) + \frac{1}{2}k'(\Delta l_0'^2 - 2x\Delta l'_0 + x^2)$$

Grouping terms:

1. Quadratic in x : $\frac{1}{2}kx^2 + \frac{1}{2}k'x^2 = \frac{1}{2}(k + k')x^2$
2. Linear in x : $x(k\Delta l_0 - k'\Delta l'_0)$. This is **zero** because $k\Delta l_0 = k'\Delta l'_0$ (Equilibrium).
3. Constant: $\frac{1}{2}k\Delta l_0^2 + \frac{1}{2}k'\Delta l_0'^2 = 0.024$ (Calculated in part 2).

Result: $ME = \frac{1}{2}Mv^2 + \frac{1}{2}(k + k')x^2 + 0.024$

c) Value of Mechanical Energy:

Initial conditions: $x_0 = -4 \text{ cm} = -0.04 \text{ m}$ and $v_0 = 0$.

$$ME = 0 + \frac{1}{2}(8 + 12)(-0.04)^2 + 0.024$$

$$ME = 10(0.0016) + 0.024 = 0.016 + 0.024$$

Result: $ME = 0.04 \text{ J}$

d) Differential Equation & Frequency:

Since friction is negligible, ME is conserved: $\frac{dME}{dt} = 0$.

$$Mvv' + (k + k')xx' = 0 \implies x'' + \frac{k + k'}{M}x = 0$$

Frequency Relation:

The natural frequency ν satisfies $\omega^2 = (2\pi\nu)^2 = \frac{k+k'}{M} = \frac{k}{M} + \frac{k'}{M}$.

Let $f = \frac{1}{2\pi} \sqrt{\frac{k}{M}}$ and $f' = \frac{1}{2\pi} \sqrt{\frac{k'}{M}}$ be the frequencies of the mass with spring (r) alone and (r') alone.

$$\text{Then } (2\pi\nu)^2 = (2\pi f)^2 + (2\pi f')^2 \implies \nu^2 = f^2 + f'^2.$$

Result:

$$v = \sqrt{f^2 + f'^2}$$

e) Solution of Differential Equation:

$$\omega^2 = \frac{20}{0.6} = \frac{100}{3} \implies \omega = \frac{10}{\sqrt{3}} \approx 5.77 \text{ rad/s.}$$

$$\text{Form: } x = X_m \cos(\omega t + \phi).$$

$$\text{At } t = 0: x = -0.04 \text{ m, } v = 0.$$

This corresponds to the negative extreme position, so $X_m = 0.04 \text{ m}$ and $\phi = \pi$.

Result:

$$x = 0.04 \cos(5.77t + \pi) \text{ (or } x = -0.04 \cos(5.77t))$$

f) Speed at $t = 2 \text{ s}$:

$$\text{Velocity } v(t) = x'(t) = -0.04(5.77) \sin(5.77t + \pi) = 0.23 \sin(5.77t).$$

$$\text{At } t = 2: v(2) = 0.23 \sin(5.77 \times 2) = 0.23 \sin(11.54 \text{ rad}).$$

$$11.54 \text{ rad is approx } 11.54 \times \frac{180}{\pi} \approx 661^\circ \equiv 301^\circ \equiv -59^\circ.$$

$$\sin(11.54) \approx -0.85.$$

$$v(2) \approx 0.23(-0.85) \approx -0.196.$$

Result:

$$v \approx 0.20 \text{ m/s}$$